

# Tejaswi Venumadhav Nerella

# Curriculum Vitae

Associate Professor, Department of Physics  
Broida Hall  
University of California, Santa Barbara  
Santa Barbara, CA 93106-9530

Contact:  
Phone: +1 (626) 826-3571  
email: teja@ucsb.edu

## Education

|                                                                                                      |           |
|------------------------------------------------------------------------------------------------------|-----------|
| <b>California Institute of Technology</b><br>Ph.D. in Physics,<br><i>Advisor:</i> Christopher Hirata | 2010-2015 |
| <b>Indian Institute of Technology, Kanpur</b><br>M.Sc (Integrated) in Physics                        | 2005-2010 |

## Academic Honors

|                                                                                                                            |                        |
|----------------------------------------------------------------------------------------------------------------------------|------------------------|
| Sloan Research Fellowship<br>Alfred P. Sloan Foundation                                                                    | 2023                   |
| Hellman Family Faculty Fellowship<br>Society of Hellman Fellows                                                            | 2023                   |
| John Bahcall Fellowship<br>Institute for Advanced Study                                                                    | 2019                   |
| Schmidt Fellowship<br>Institute for Advanced Study                                                                         | 2015 - 2018            |
| Robert A. Millikan Fellowship<br>California Institute of Technology                                                        | 2010                   |
| International Fulbright Science and Technology Award<br>Bureau of Education and Cultural Affairs, U.S. Department of State | 2010                   |
| President's Gold Medal for the best academic performance in<br>the graduating class in all disciplines, IIT Kanpur         | 2010                   |
| General Proficiency Medal for the best academic performance in<br>the graduating class in Physics, IIT Kanpur              | 2010                   |
| Summer Undergraduate Research Fellowship<br>California Institute of Technology                                             | 2007, 2008             |
| Academic Excellence Award<br>IIT Kanpur                                                                                    | 2007, 2008, 2009, 2010 |
| Silver Medal, 36th International Physics Olympiad                                                                          | 2005                   |
| KVPY Fellowship<br>Department of Science and Technology, Govt. of India                                                    | 2004                   |
| NTSE Fellowship<br>National Council of Educational Research and Training, Govt. of India                                   | 2003                   |

## Work Experience

---

|                                                                                                                             |                   |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Associate Professor</b><br>University of California, Santa Barbara                                                       | Jul 2026-present  |
| <b>Assistant Professor</b><br>University of California, Santa Barbara                                                       | Jul 2020-Jul 2026 |
| <b>Visiting Professor</b><br>International Center for Theoretical Sciences, Bangalore                                       | 2020-present      |
| <b>Member</b><br>Institute for Advanced Study, Princeton                                                                    | Sep 2015-Jul 2020 |
| <b>Associate</b><br>International Center for Theoretical Sciences, Bangalore                                                | 2019-2020         |
| <b>Graduate Student</b><br>California Institute of Technology, Pasadena<br><i>Advisor:</i> Christopher M. Hirata            | Sep 2010-Aug 2015 |
| <b>Visiting Scientist</b><br>Max-Planck-Institut für Physik komplexer Systeme, Dresden<br><i>Advisor:</i> Roderich Moessner | May-August 2009   |
| <b>Summer Undergraduate Research Fellow</b><br>California Institute of Technology, Pasadena<br><i>Advisor:</i> Re'em Sari   | May-August 2008   |
| <b>Summer Undergraduate Research Fellow</b><br>California Institute of Technology, Pasadena<br><i>Advisor:</i> Andrew Lange | May-August 2007   |

## Current and Pending Support

---

|                         |                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------|
| Project/Proposal Title: | WoU-MMA: Expanding the Horizons of Gravitational Wave Searches and Parameter Estimation |
| Source of Support:      | NSF                                                                                     |
| Award Number:           | 2012086                                                                                 |
| Project Location:       | University of California, Santa Barbara                                                 |
| Total Amount:           | \$300000                                                                                |
| Starting/Ending Dates:  | 07/01/2020–06/30/2024                                                                   |
| Person-months Per Year: | 4 calendar months/year                                                                  |

|                         |                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------|
| Project/Proposal Title: | WoU-MMA: Targeted Search for Binary Mergers with Multiple Harmonics in Gravitational Wave Data |
| Source of Support:      | NSF                                                                                            |
| Award Number:           | 2309360                                                                                        |
| Project Location:       | University of California, Santa Barbara                                                        |
| Total Amount:           | \$300000                                                                                       |
| Starting/Ending Dates:  | 05/01/2023–04/30/2026                                                                          |
| Person-months Per Year: | 4 calendar months/year                                                                         |

|                         |                                                          |
|-------------------------|----------------------------------------------------------|
| Project/Proposal Title: | Searching for Missing Links in the Black Hole Population |
| Source of Support:      | Alfred P. Sloan Foundation                               |
| Project Location:       | University of California, Santa Barbara                  |
| Total Amount:           | \$75000                                                  |
| Starting/Ending Dates:  | 09/15/2023–09/15/2025                                    |
| Person-months Per Year: | 2 calendar months/year                                   |

  

|                         |                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------|
| Project/Proposal Title: | Cosmological recombination on small-scales, and implications for the Hubble constant |
| Source of Support:      | Society of Hellman Fellows                                                           |
| Project Location:       | University of California, Santa Barbara                                              |
| Total Amount:           | \$24800                                                                              |
| Starting/Ending Dates:  | 06/01/2023–08/31/2024                                                                |
| Person-months Per Year: | 2 calendar months/year                                                               |

  

|                         |                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------|
| Project/Proposal Title: | IAS Gravitational Wave Virtual Center                                             |
| Source of Support:      | Jonathan M. Nelson Center for Collaborative Research                              |
| Project Location:       | Institute for Advanced Study                                                      |
| Total Amount:           | \$300000                                                                          |
| Starting/Ending Dates:  | 07/01/2026–07/01/2029                                                             |
| Person-months Per Year: | 3 (Co-Investigator, travel support, and partial postdoc support)                  |
| URL:                    | <a href="https://www.ias.edu/nccr/projects">https://www.ias.edu/nccr/projects</a> |

## Graduate Committees

---

| Student Name              | Degree Completed | Role   |
|---------------------------|------------------|--------|
| Anantpurkar, Isha         | In progress      | Chair  |
| Schiff, Jonathan          | In progress      | Chair  |
| Garza, Sierra             | In progress      | Chair  |
| Kwon, James (Kyubin)      | In progress      | Chair  |
| Becerra Espinoza, Joaquin | In progress      | Chair  |
| Surgent, William          | In progress      | Chair  |
| Huang, Jiamu              | In progress      | Member |
| Gonzalez-Hernandez, Diego | In progress      | Member |
| Clay, Hawkins             | In progress      | Member |
| Kim, Crystal              | In progress      | Member |
| Breckenridge, Josh        | In progress      | Member |
| Vancil, Connor            | In progress      | Member |
| Palatnick, Skyler         | In progress      | Member |
| Liang, Naixin             | In progress      | Member |

## Undergraduate Students Supervised

---

| Name         | Years     | Co-Mentor            | Project Title                                                                 | Position Project                  | After           |
|--------------|-----------|----------------------|-------------------------------------------------------------------------------|-----------------------------------|-----------------|
| Liu, Cuishan | 2021–2022 | Javier Roulet (KITP) | Coordinates for Gravitational Wave Parameter Estimation using Fisher Matrices | Graduate (University of Virginia) | Student of Vir- |

|                  |              |                                 |                                                                                                |                                                      |
|------------------|--------------|---------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------|
| Carrel, Dashiell | 2021–2023    | Jennifer Barnes (KITP)          | AT2017gfo-like Photometry From a Single Component of Binary Neutron Star Merger Ejecta         | Graduate Student (University of California Berkeley) |
| Wang, Jiayu      | 2022–2024    | Ajit Kumar Mehta (UCSB postdoc) | Improving the Detection Sensitivity of Gravitational-Wave Searches in Non-Stationary Data      |                                                      |
| Wei, Fansen      | 2022–2023    |                                 | Calibration Uncertainty in Gravitational-Wave Parameter Estimation                             | Graduate Student (Brown University)                  |
| Xu, Junyan       | 2023–2025    |                                 | Parameter Estimation of Long-Duration Signals in Third-Generation Gravitational Wave Detectors | Graduate Student (University of Minnesota)           |
| Ding, Yangxiao   | 2025–Present | James Kwon                      | Interface modes in neutron stars                                                               |                                                      |
| Hahn, Christian  | 2026–Present | Tousif Islam                    | TBD                                                                                            |                                                      |

## Postdoctoral Scholars Supervised

| Name             | Years        | Note        | Position After UCSB                                   |
|------------------|--------------|-------------|-------------------------------------------------------|
| Roulet, Javier   | 2021–2022    | KITP Fellow | Sherman Fairchild Postdoctoral Scholar (Caltech)      |
| Barnes, Jennifer | 2021–2023    | KITP Fellow | Senior Algorithm Engineer, 3M, Minneapolis, Minnesota |
| Yu, Hang         | 2022–2023    | KITP Fellow | Assistant Professor, Montana State University         |
| Ajit Kumar Mehta | 2022–2025    |             | Assistant Professor, Chennai Mathematical Institute   |
| Tousif Islam     | 2023–Present | KITP Fellow |                                                       |

## Other Supervision/Mentoring

| Name          | Years     | Note                 | Position After Supervision |
|---------------|-----------|----------------------|----------------------------|
| Islam, Tousif | 2022–2023 | KITP Graduate Fellow | KITP Fellow                |

## University Service

---

| Name of Committee/Organization                          | Period    | Position |
|---------------------------------------------------------|-----------|----------|
| Graduate Admissions, Physics Department                 | 2021-2023 | Member   |
| Winter quarter Colloquium Committee, Physics Department | 2021-2022 | Co-Chair |
| Spring quarter Colloquium Committee, Physics Department | 2022-2023 | Co-Chair |
| Student Awards, Physics Department                      | 2024-2025 | Chair    |
| Freshman Advisor, Physics Department                    | 2025-2026 | Advisor  |
| Undergraduate Research, Physics Department              | 2025-2026 | Member   |

## Professional Service

---

- Referee for the Astrophysical Journal Letters
- Referee for Astroparticle Physics
- Referee for the Astrophysical Journal
- Referee for Monthly Notices of the Royal Astronomical Society Letters
- Referee for Monthly Notices of the Royal Astronomical Society
- Referee for Physical Review D
- Panel reviewer for ERC, BSF, NSF
- Organizer of Prospects in Theoretical Physics 2025 program on *Gravitational Waves from Theory to Observation* URL: <https://www.ias.edu/pitp-2025-details>
- *Organizer of the 2025 KITP Program: Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys* URL: <https://www.kitp.ucsb.edu/activities/stellarbh25>
- *Organizer of the 2025 KITP Conference: The Lifecycle of Stellar Black Holes* URL: <https://www.kitp.ucsb.edu/activities/stellarbh-c25>

## Publication Highlights

---

**Total h-index: 32**

**Top five cited published papers (not counting  $n^{\text{th}}$  author papers):**

1. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030  
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo  
Citation count: 378
2. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009  
Title: New binary black hole mergers in the LIGO-Virgo O3a data  
Citation count: 207
3. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007

Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run

Citation count: 199

4. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011

Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO

Citation count: 191

5. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030

Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

Citation count: 156

## Refereed Publications

---

1. Perera, A., Zackay, B., **Venumadhav, T.**, (2026), Monthly Notices of the Royal Astronomical Society, 551, stag1592

Title: Expanding the population of short gamma-ray transients with a coherent fermi/GBM search: a 13-yr catalogue of short GRBs

2. Islam, T., **Venumadhav, T.**, Wadekar, D., (2026), Physical Review Letters, 137, 031404

Title: Progenitor of the Recoiling Supermassive Black Hole RBH-1 Identified Using HST and JWST Imaging

3. Maity, K.N., Jana, S., **Venumadhav, T.**, Barsode, A., Ajith, P., (2026), Physical Review D, 113, 103522

Title: Strong lensing cosmography using binary-black-hole mergers: Prospects for the near future

4. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), Physical Review Letters, 136, 071401

Title: Relativistic and Dynamical Love Numbers

5. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2026), Physical Review D, 113, 023003

Title: Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA

6. Mehta, A.K., Wadekar, D., Anantpurkar, I., Roulet, J., **Venumadhav, T.**, Islam, T., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 124023

Title: Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline

7. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 112, 104039

Title: Post-Newtonian theory-inspired framework for characterizing eccentricity in gravitational waveforms

8. Perera, A., Zackay, B., **Venumadhav, T.**, (2025), The Astrophysical Journal Supplement Series, 281, 4

Title: A New Search Pipeline for Short Gamma-Ray Bursts in Fermi/GBM Data—A 50% Increase in the Number of Detections

9. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M., (2025), Physical Review D, 112, 104025

Title: Sampler-free gravitational wave inference using matrix multiplication

10. Jana, S., Kapadia, S.J., **Venumadhav, T.**, More, S., Ajith, P., (2025), Physical Review Letters, 135, 111402

Title: Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes

11. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 044070  
Title: Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms
12. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 111, L081503  
Title: Universal phenomenological relations between spherical harmonic modes in nonprecessing eccentric binary black hole merger waveforms
13. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 111, 024049  
Title: New binary black hole mergers in the LIGO-Virgo O3b data
14. Jana, S., J Kapadia, S., **Venumadhav, T.**, More, S., Ajith, P., (2024), Classical and Quantum Gravity, 41, 245010  
Title: Strong-lensing cosmography using third-generation gravitational-wave detectors
15. Wadekar, D., **Venumadhav, T.**, Mehta, A.K., Roulet, J., Olsen, S., Mushkin, J., Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 084035  
Title: New approach to template banks of gravitational waves with higher harmonics: Reducing matched-filtering cost by over an order of magnitude
16. Chia, H.S., Edwards, T.D.P., Wadekar, D., Zimmerman, A., Olsen, S., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 063007  
Title: In pursuit of Love numbers: First templated search for compact objects with large tidal deformabilities in the LIGO-Virgo data
17. Roulet, J., Mushkin, J., Wadekar, D., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 044010  
Title: Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation, and sky localization
18. Wadekar, D., **Venumadhav, T.**, Roulet, J., Mehta, A.K., Zackay, B., Mushkin, J., Zaldarriaga, M., (2024), Physical Review D, 110, 044063  
Title: New search pipeline for gravitational waves with higher-order modes using mode-by-mode filtering
19. Roulet, J., **Venumadhav, T.**, (2024), Annual Review of Nuclear and Particle Science, 74, 207  
Title: Inferring Binary Properties from Gravitational-Wave Signals
20. Yu, H., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2023), Physical Review D, 108, 064059  
Title: Accurate and efficient waveform model for precessing binary black holes
21. Jana, S., Kapadia, S.J., **Venumadhav, T.**, Ajith, P., (2023), Physical Review Letters, 130, 261401  
Title: Cosmography Using Strongly Lensed Gravitational Waves from Binary Black Holes
22. Yu, H., Weinberg, N.N., Arras, P., Kwon, J., **Venumadhav, T.**, (2023), Monthly Notices of the Royal Astronomical Society, 519, 4325  
Title: Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals
23. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 123015  
Title: Removing degeneracy and multimodality in gravitational wave source parameters
24. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009  
Title: New binary black hole mergers in the LIGO-Virgo O3a data
25. Chia, H.S., Olsen, S., Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 024009  
Title: Signs of higher multipoles and orbital precession in GW151226

26. Olsen, S., Roulet, J., Chia, H.S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083036  
Title: Mapping the likelihood of GW190521 with diverse mass and spin priors
27. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083010  
Title: Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs
28. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M., (2021), Physical Review D, 104, 063034  
Title: Detecting gravitational waves in data with non-stationary and non-Gaussian noise
29. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030  
Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers
30. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 123022  
Title: Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events
31. Huang, Y., Haster, C.J., Roulet, J., Vitale, S., Zimmerman, A., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 103024  
Title: Source properties of the lowest signal-to-noise-ratio binary black hole detections
32. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030  
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo
33. Dai, L., Kaurov, A.A., Sharon, K., Florian, M., Miralda-Escudé, J., **Venumadhav, T.**, Frye, B., Rigby, J.R., Bayliss, M., (2020), Monthly Notices of the Royal Astronomical Society, 495, 3192  
Title: Asymmetric surface brightness structure of caustic crossing arc in SDSS J1226+2152: a case for dark matter substructure
34. Samsing, J., **Venumadhav, T.**, Dai, L., Martinez, I., Batta, A., Lopez, M., Ramirez-Ruiz, E., Kremer, K., (2019), Physical Review D, 100, 043009  
Title: Probing the black hole merger history in clusters using stellar tidal disruptions
35. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007  
Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
36. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011  
Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO
37. Kaurov, A.A., Dai, L., **Venumadhav, T.**, Miralda-Escudé, J., Frye, B., (2019), The Astrophysical Journal, 880, 58  
Title: Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403
38. Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2019), Physical Review D, 99, 123022  
Title: Template bank for compact binary coalescence searches in gravitational wave data: A general geometric placement algorithm
39. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M., (2018), Physical Review D, 98, 103513  
Title: Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn



40. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escudé, J., (2018), The Astrophysical Journal, 867, 24  
Title: Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars
41. Kaurov, A.A., **Venumadhav, T.**, Dai, L., Zaldarriaga, M., (2018), The Astrophysical Journal Letters, 864, L15  
Title: Implication of the Shape of the EDGES Signal for the 21 cm Power Spectrum
42. Hirata, C.M., Mishra, A., **Venumadhav, T.**, (2018), Physical Review D, 97, 103521  
Title: Detecting primordial gravitational waves with circular polarization of the redshifted 21 cm line. I. Formalism
43. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J., (2017), The Astrophysical Journal, 850, 49  
Title: Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters
44. **Venumadhav, T.**, Oklopčić, A., Gluscevic, V., Mishra, A., Hirata, C.M., (2017), Physical Review D, 95, 083010  
Title: New probe of magnetic fields in the preionization epoch. I. Formalism
45. Gluscevic, V., **Venumadhav, T.**, Fang, X., Hirata, C., Oklopčić, A., Mishra, A., (2017), Physical Review D, 95, 083011  
Title: New probe of magnetic fields in the pre-reionization epoch. II. Detectability
46. Dai, L., **Venumadhav, T.**, Sigurdson, K., (2017), Physical Review D, 95, 044011  
Title: Effect of lensing magnification on the apparent distribution of black hole mergers
47. **Venumadhav, T.**, Chang, T.C., Doré, O., Hirata, C.M., (2016), The Astrophysical Journal, 826, 116  
Title: A Practical Theorem on Using Interferometry to Measure the Global 21-cm Signal
48. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M., (2016), Physical Review D, 94, 043515  
Title: Sterile neutrino dark matter: Weak interactions in the strong coupling epoch
49. **Venumadhav, T.**, Hirata, C., (2015), Physical Review D, 91, 123009  
Title: Stability of small-scale baryon perturbations during cosmological recombination
50. **Venumadhav, T.**, Zimmerman, A., Hirata, C.M., (2014), The Astrophysical Journal, 781, 23  
Title: The Stability of Tidally Deformed Neutron Stars to Three- and Four-mode Coupling
51. Venumadhav, T., Haque, M., Moessner, R., (2010), Physical Review B, 81, 054305  
Title: Finite-rate quenches of site bias in the Bose-Hubbard dimer

## Preprints on the Arxiv

---

1. Maity, K.N., Jana, S., Barsode, A., **Venumadhav, T.**, Ajith, P., (2026), arXiv:2609.05020  
Title: Probing Dark Matter with Strongly Lensed Binary Black Hole Mergers: Prospects in the Near Future
2. Ho-Yeuk Cheung, M., Wadekar, D., Zaldarriaga, M., **Venumadhav, T.**, Roulet, J., Mehta, A.K., (2026), arXiv:2607.21834  
Title: The diffraction-lensing interpretation of GW231123 with astrophysical priors
3. Yu, H., Nicolini, G., Lau, S.Y., Kwon, K.J., **Venumadhav, T.**, Andersson, N., Pnigouras, P., Gittins, F., Nanda, A., (2026), arXiv:2607.07943  
Title: Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions
4. Abhishek Hegade K., R., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), arXiv:2605.08569  
Title: The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response
5. Islam, T., Wadekar, D., **Venumadhav, T.**, Zaldarriaga, M., Mehta, A.K., Roulet, J., Zackay, B., (2026), arXiv:2605.11280  
Title: Discovery of Interpretable Surrogates via Agentic AI: Application to Gravitational Waves

6. Islam, T., **Venumadhav, T.**, Wadekar, D., Mehta, A.K., Roulet, J., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2604.07388  
Title: GW190711\_030756 and GW200114\_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog
7. Zhou, Z., Wadekar, D., Roulet, J., Ivashtenko, O., **Venumadhav, T.**, Islam, T., Mehta, A.K., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2603.05784  
Title: Searching for precessing binary systems with mode-by-mode filtering and marginalization
8. Islam, T., **Venumadhav, T.**, Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2509.20556  
Title: Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers
9. Wadekar, D., Pimpalkar, A., Ho-Yeuk Cheung, M., Wandelt, B., Berti, E., Mehta, A.K., **Venumadhav, T.**, Roulet, J., Islam, T., Zackay, B., Mushkin, J., Zaldarriaga, M., (2025), arXiv:2507.08318  
Title: Improving gravitational wave search sensitivity with TIER: Trigger Inference using Extended strain Representation
10. Schiff, J., **Venumadhav, T.**, (2025), arXiv:2506.16517  
Title: Primordial magnetic fields and modified recombination histories
11. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2504.12420  
Title: gwharmon: first data-driven surrogate for eccentric harmonics in binary black hole merger waveforms
12. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2025), arXiv:2503.11837  
Title: Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of  $g$ -modes in merging neutron star binaries
13. Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., Islam, T., (2025), arXiv:2501.17939  
Title: Significant increase in sensitive volume of a gravitational wave search upon including higher harmonics
14. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2024), arXiv:2410.03831  
Title: Resonance Locking of Anharmonic  $g$ -Modes in Coalescing Neutron Star Binaries
15. Wadekar, D., Roulet, J., **Venumadhav, T.**, Mehta, A.K., Zackay, B., Mushkin, J., Olsen, S., Zaldarriaga, M., (2023), arXiv:2312.06631  
Title: New black hole mergers in the LIGO-Virgo O3 data from a gravitational wave search including higher-order harmonics
16. Islam, T., Roulet, J., **Venumadhav, T.**, (2022), arXiv:2210.16278  
Title: Factorized Parameter Estimation for Real-Time Gravitational Wave Inference
17. Dai, L., Zackay, B., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2020), arXiv:2007.12709  
Title: Search for Lensed Gravitational Waves Including Morse Phase Information: An Intriguing Candidate in O2
18. Coleman, M.S.B., **Venumadhav, T.**, Zackay, B., (2019), arXiv:1903.04978  
Title: Gravitational-wave-moderated Accretion: The Case of ES Ceti
19. Haris, K., Mehta, A.K., Kumar, S., **Venumadhav, T.**, Ajith, P., (2018), arXiv:1807.07062  
Title: Identifying strongly lensed gravitational wave signals from binary black hole mergers
20. Zackay, B., Dai, L., **Venumadhav, T.**, (2018), arXiv:1806.08792  
Title: Relative Binning and Fast Likelihood Evaluation for Gravitational Wave Parameter Estimation
21. Dai, L., **Venumadhav, T.**, Zackay, B., (2018), arXiv:1806.08793  
Title: Parameter Estimation for GW170817 using Relative Binning
22. Dai, L., **Venumadhav, T.**, (2017), arXiv:1702.04724  
Title: On the waveforms of gravitationally lensed gravitational waves

## N-th Author Papers

---

1. Raaijmakers, G., et. al., (2021), The Astrophysical Journal, 922, 269  
Title: The Challenges Ahead for Multimessenger Analyses of Gravitational Waves and Kilonova: A Case Study on GW190425
2. Weltman, A., et. al., (2020), Publications of the Astronomical Society of Australia, 37, e002  
Title: Fundamental physics with the Square Kilometre Array
3. Doré, O., et. al., (2014), arXiv e-prints  
Title: Cosmology with the SPHEREX All-Sky Spectral Survey

## Talks and Presentations

---

1. Invited theory colloquium, Tata Institute of Fundamental Research, Mumbai. 2025
2. Invited talk, Current Themes in Astrophysics and Particle Physics, NBIA, Copenhagen. 2025
3. Invited seminar, International Center for Theoretical Sciences, Bangalore. 2025
4. Invited review talk, APS Global Physics Summit 2025, Anaheim. 2025
5. Invited seminar, Weinberg Institute, University of Texas, Austin. 2025
6. Invited seminar, City University of Hong Kong, Hong Kong. 2025
7. Invited plenary talk, IAS Program on Fundamental Physics, HKUST, Hong Kong. 2025
8. Invited DGRAV seminar, APS DGRAV (online). 2024
9. Invited theory colloquium, Tata Institute of Fundamental Research, Mumbai. 2024
10. Invited talk, 21st Century Cosmology: tensions, anomaly and new physics, Ashoka University, Sonapat. 2024
11. Invited talk, XGMDC, Penn State University, PA. 2024
12. Invited seminar, IIT Hyderabad, Hyderabad. 2023
13. Invited talk, RESCEU-NBI Workshop on Gravitational-Wave Sources, Tokyo. 2023
14. Invited talk, International Conference on Gravitation and Cosmology (ICGC), Guwahati. 2023
15. Invited talk, Gravitational waves in the desert, Ein Gedi, Israel<sup>1</sup>. 2023
16. Invited talk, Gravitational Wave Physics and Astronomy Workshop, Melbourne. 2022
17. Invited Astrophysics seminar, International Center for Theoretical Sciences, Bangalore. 2022
18. Invited Astrophysics seminar, Raman Research Institute, Bangalore. 2022
19. Invited Astrophysics seminar, Indian Institute of Science, Bangalore. 2022
20. Invited talk at the ISSI (International Space Science Institute) Workshop on Strong Gravitational Lensing (virtual). 2022
21. Invited talk at the SRITP workshop on "EM counterparts to GW sources" at the Weizmann Institute, Rehovot. 2022
22. Invited talk at the KITP conference titled "Storming the Gravitational Wave Frontier". 2022
23. Invited Panelist at the APS April Meeting on the panel Data Analysis in Astrophysics (virtual). 2021
24. Invited Astrophysics Colloquium, Massachusetts Institute of Technology. 2020
25. Invited Cosmology seminar (virtual), CERN. 2020
26. Invited Seminar, CITA, Toronto. 2020
27. Invited Talk, Gravitational wave searches and parameter estimation in the era of detections, Schloss Ringberg. 2020
28. Invited Seminar, Indian Institute of Technology, Mumbai. 2020

---

<sup>1</sup>Postponed

29. Invited Seminar, Tata Institute of Fundamental Research, Mumbai. 2019
30. Invited Talk, Frank N. Bash Symposium, UT Austin. 2019
31. Talk, Gravitational Wave Physics and Astronomy Workshop, Tokyo. 2019
32. Invited Talk, Black Holes and Neutron Stars with Gravitational Waves, YITP, Kyoto. 2019
33. Invited Colloquium, Black Hole Initiative, Harvard. 2019
34. Invited panelist, The Future of Gravitational-Wave Astronomy, Bangalore. 2019
35. Invited Seminar, International Centre for Theoretical Sciences, TIFR. 2019
36. Invited Seminar, Princeton Gravity Initiative, Princeton. 2019
37. Invited Seminar, Albert Einstein Institute, Potsdam. 2019
38. Invited Seminar, Center for Cosmology and Particle Physics, NYU. 2019
39. Invited Seminar, Astronomy and Astrophysics, UC Santa Barbara. 2019
40. Invited Colloquium, Department of Physics, UC Santa Barbara. 2019
41. Invited panelist, Physics and Astrophysics at the eXtreme, IUCAA, Pune. 2018
42. Invited talk, Thermal history of the Universe at intermediate redshift: progress with 21cm absorption measurements, CERN. 2018
43. Talk, Shedding Light on the Dark Universe with Extremely Large Telescopes, UCLA. 2018
44. Invited Cosmology seminar, JHU, Baltimore. 2017
45. Invited Seminar, CITA, Toronto. 2017
46. Talk, Fundamental Physics with the Square Kilometer Array, Mauritius. 2017
47. Invited talk, Tianlai Collaboration Meeting, Fermilab, Batavia. 2016
48. Invited talk, CMB Spectral Distortions From Cosmic Baryon Evolution, RRI, Bengaluru. 2016
49. Invited seminar, International Centre for Theoretical Sciences, TIFR. 2016
50. Invited cosmology seminar, Perimeter institute. 2016
51. Cosmology lunch, joint w/ IAS and Princeton University. 2016
52. Astrophysics informal seminar, IAS. 2016
53. Seminar, Inter University Center for Astronomy and Astrophysics, Pune. 2015
54. Seminar, National Center for Radio Astronomy, Pune. 2015
55. Talk, The Primordial Universe after Planck, IAP, Paris. 2014
56. Seminar, McGill University, Montreal. 2014
57. Seminar, CITA, Toronto. 2014
58. ITC Seminar, Harvard University, Boston. 2014
59. Cosmology lunch, joint w/ IAS and Princeton University. 2014
60. Talk, Theoretical Astrophysics in Southern California (TASC), UCSD, San Diego. 2014
61. Special seminar, KICP, University of Chicago. 2014
62. Cosmology Lunch talk, CCAPP, Ohio State University, Columbus. 2014
63. Poster, Gravitational Wave Physics and Astronomy Workshop (GWPAW) at IUCAA, Pune. 2013
64. Seminar, Inter University Center for Astronomy and Astrophysics, Pune. 2013
65. Talk, Theoretical Astrophysics in Southern California (TASC), Carnegie Observatories, Pasadena. 2012
66. Poster, Summer school on cosmology, ICTP, Trieste. 2012

## References

---

Matias Zaldarriaga  
Richard Black Professor  
Institute for Advanced Study  
1 Einstein Drive  
Princeton, NJ 08540, USA  
*email:* matiasz@ias.edu

Bangalore Sathyaprakash  
Elsbach Professor of Physics  
Department of Astronomy and Astrophysics  
Penn State University  
525 Davey Laboratory  
University Park, PA 16802, USA  
*email:* bss25@psu.edu

Jordi Miralda Escudé  
Institut de Ciències del Cosmos  
Universitat de Barcelona  
08028 Barcelona Catalonia, Spain  
*email:* miralda@icc.ub.edu

Marc Kamionkowski  
William R. Kenan, Jr. Professor of Physics and  
Astronomy  
Johns Hopkins University  
Department of Physics and Astronomy  
Bloomberg 439  
3400 North Charles Street  
Baltimore, MD 21210, USA  
*email:* kamion@jhu.edu

Alessandra Buonanno  
Director, Max Planck Institute for Gravitational  
Physics  
Max-Planck-Institut für Gravitationsphysik (Albert-  
Einstein-Institut)  
Am Mühlenberg 1  
D-14476 Potsdam, Germany  
*email:* alessandra.buonanno@aei.mpg.de

Christopher M. Hirata  
The Ohio State University  
191 West Woodruff Lane  
Columbus, OH 43210, USA  
*email:* hirata.10@osu.edu